

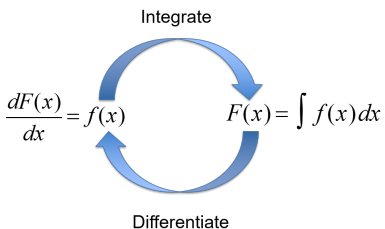
EART11200 - The Natural Scientist's Toolkit
Mathematics and Physics

Week 7 Lecture - Integration using Newton's Laws

Last Week...

- Integration is the **reverse** of differentiation.
- Since the derivative function provides us with a gradient, integration therefore provides us with a function that can be used to calculate absolute values.
- The $\int$ symbol was introduced by Gottfried Wilhelm Leibniz in the 17th century to represent summation of infinitely many infinitesimally small quantities.

e.g. An equation that calculates the rate of change of temperature in the past, present or future



e.g. An equation that calculates the temperature in the past, present or future

Last Week...

- We can reverse some common derivatives, we arrive at the following functions.
- The more you practice these procedures, the easier it will become to apply them without looking them up.
- Use your math's survival pack when needed!

Derivative $\frac{dy}{dx}$	**Integral** y
0	d
constant	$constant \times x + d$
x	$\frac{x^2}{2} + d$
x^n	$\frac{x^{n+1}}{n+1} + d$
e^x	$e^x + d$
ke^{kx}	$e^{kx} + d$
$\frac{1}{x}$	$ln(x) + d$
$cos(x)$	$sin(x) + d$
$sin(x)$	$-cos(x) + d$

Last Week...

- **Indefinite integration** (i.e. without limits)

$$\int 3x^2\,dx = x^3 + d$$

- **Definite integration** (i.e. with limits)

$$\int_0^1 3x^2\,dx = \left[x^3\right]_0^1 = 1^3 - 0^3 = 1$$

Derivative $\frac{dy}{dx}$	**Integral** y
0	d
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$\frac{1}{x}$	$ln(x) + d$
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Last Week...

- The equation describing radioactive decay is:

$$\frac{dN}{dt} = -\lambda N$$

where N is the number of unstable nuclei, t is time and λ is called the decay rate.

- This is a differential equation, and we used **separation of variables** to show that it has solution:

$$N(t) = N_0 \exp(-\lambda t)$$

where N_0 is the total number of nuclei in the sample at $t = 0$.

Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

(1) Understand and describe Newton's Laws of Motion to explain the relationship between force, mass, and acceleration in physical systems.

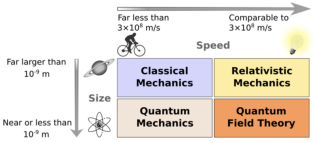
(2) Apply Newton's Second Law to formulate ordinary differential equations (ODEs) that model the motion of objects in various contexts.

(3) Use integration to solve ODEs and determine physical behaviour in real-world problems.

Newton's Laws

- Before the 20th century, our understanding of the physical world was built entirely on classical mechanics. More specifically - **Newtonian Physics**.
- This framework emerged from the collective efforts of the "Scientific Revolution".
- Still an active area of theoretical and experimental physics.
- In the `Principia`, Newton formulated **three laws** to explain planetary motion, in order to explain why the orbits of the planets are ellipses rather than circles.
- These laws extend far beyond planetary motion, offering a framework for understanding force and motion.






NewScientist

Physics

Is space-time quantum? Six ways to unpick the fabric of the universe

One of the biggest questions in physics asks whether space-time is classical or quantum in nature. From slow neutrinos to quantum foam, these experiments are hoping to finally answer it

By Lyndie Chiou

25 October 2023

Newton's Laws

- **Law of Inertia:** A body at rest or moving at a constant speed in a straight line will remain in that state unless acted upon by an external force.
- **Law of Force and Momentum:** The rate of change of a body's momentum is directly proportional to the force applied, with the change occurring in the same direction as the force (i.e. if a force acts on a body, that body will experience a change in momentum).
- **Law of Action and Reaction:** When two bodies interact, they exert forces on each other that are equal in magnitude and opposite in direction. You will have likely come across this law before; also known as the law of action and reaction.

Newton's Second Law

- In this course, we focus on Newton's Second Law, which is given by the following equation:

$$F = ma$$

  where F is the net force acting on a body that has a mass m and therefore acceleration a.
- Acceleration of an object depends on the mass of the object and the amount of force applied.
- In Week 5, we discussed second-order derivatives and observed that acceleration (a) is the rate of change of velocity (v), while velocity (v) is the rate of change of position or distance (x).
- Newton's second law can therefore be expressed as ordinary differential equations:

$$F = m\,\frac{dv}{dt} \qquad F = m\,\frac{d}{dt}\left(\frac{dx}{dt}\right) \qquad F = m\,\frac{d^2x}{dt^2} \qquad F = mv\,\frac{dv}{dx}$$

Newton's Second Law

- To help understand how this works, imagine a plane moving through the atmosphere.
- The plane has a constant mass m and its momentum depends on its velocity V, which is controlled solely by the plane (i.e., there are no external forces acting on it).
- The change in velocity between any two time points is $m(V_1 - V_0)$. If we define force as the rate of change of momentum with respect to time, then:

$$F = m\frac{V_1 - V_0}{t_1 - t_0}$$

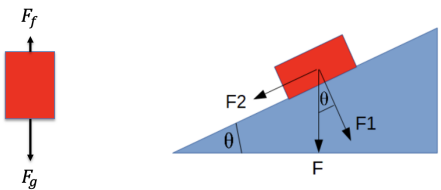
- This resembles the calculation of a gradient from Week 1!

Newton's Second Law

- As seen in Chapters 2, taking the limit as $t_1 - t_0 \to 0$, this becomes us the derivative:

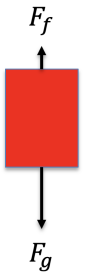
$$F = m\,\frac{dv}{dt}$$

- By formulating the total forces acting on an object, we can then solve the ODE to as a function of time to tell us something about the processes.
- How do we determine the forces acting on a body, thus write our ODE?



Example: Ball falling in a vacuum

- Consider an object falling in a vacuum.
- The object experiences only Earth's gravitational force, which acts vertically from the center of the object toward the Earth's center.
- Therefore we see that $F_f = 0$ and $F_g = mg$ where $g = 9.8$ is the gravitational constant.
- If we assume an object is falling in a vacuum, can we prove that a brick fall the same distance as a feather in the same amount of time? Yes!



Example: Ball falling in a vacuum

- We add up all the forces in the direction of travel and minus the forces in the opposite direction:

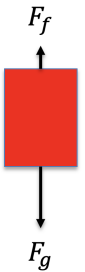
$$F = mg$$

- This allows us to model the motion using a first-order ordinary differential equation (ODE) for velocity:

$$mg = m\,\frac{dv}{dt}$$

- Canceling the mass (m) gives:

$$\frac{dv}{dt} = g$$

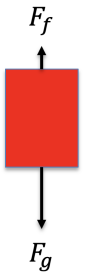


Example: Ball falling in a vacuum

- Using separation of variables, we can re-write our expression as an integral so we can calculate velocity as a function of time.
- We integrate to find the velocity, assuming the object starts from rest ($v = 0$) at $t = 0$:

$$\int_0^v dv = \int_0^t g\, dt \quad \Longrightarrow \quad [v]_0^v = [gt]_0^t \quad \Longrightarrow \quad v(t) = gt$$

- Thus, the velocity of any object in a vacuum increases linearly with time, regardless of its mass, therefore, all objects fall at the same velocity with respect to time.



Example: Ball falling in a vacuum

- However, we want to show that all objects fall the same distance with respect to time. Remember that, velocity is the rate of change for the distance so we get the following ODE

$$v(t) = \frac{dx}{dt} = gt$$

- Integrating again, and assuming the object starts from $x = 0$ at $t = 0$, we get:

$$\int_0^x dx = \int_0^t gt\, dt \implies [x]_0^x = \left[\frac{1}{2}gt\right]_0^t \implies x(t) = \frac{1}{2}gt^2$$

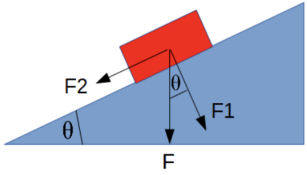
- This shows that all objects fall the same distance (quadratically) over the same amount of time in a vacuum, regardless of their mass.



Example: Object on an Incline

- Consider an object resting on an incline.
- The object experiences only Earth's gravitational force, which acts vertically from the center of the object toward the Earth's center (i.e. $F_g = mg$)
- The forces can be resolved into two components:
  - the force acting perpendicular to the slope ($F1$).
  - the force dragging the block down the slope ($F2$).
- What is the magnitude of both $F1$ and $F2$?
- Using trigonometry, we can write these as follows:

$$F1 = mg\cos(\theta) \qquad F2 = mg\sin(\theta)$$



Example: Object on an Incline

- Assuming that there is no friction acting against the object, we expect the block to travel down the incline.
- We add up all the forces in the direction of travel and minus the forces in the opposite direction:

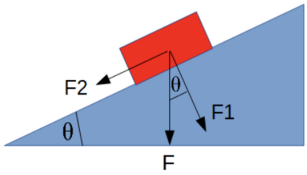
$$F = mg\sin(\theta)$$

- This means the force $F2$ can be used to formulate our ODE:

$$m\frac{dv}{dt} = mg\sin(\theta)$$

- Canceling the mass (m) gives:

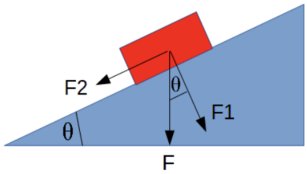
$$\frac{dv}{dt} = g\sin(\theta)$$



Example: Object on an Incline

- Using separation of variables, we can integrate and calculate the velocity as a function of time. We integrate to find the velocity, assuming the object starts from rest ($v = 0$) at $t = 0$:

$$\int_0^v dv = \int_0^t g\sin(\theta)\, dt \quad \Longrightarrow \quad [v]_0^v = [gt\sin(\theta)]_0^t$$
$$\Longrightarrow \quad v(t) = gt\sin(\theta)$$



- Thus, the velocity of an object on an incline increases linearly with time.
- The rate of increase depends on the incline's angle (θ) but is independent of the object's mass.

Newton's Second Law

- Both of these examples are simplified.
  - Could we derive a general expression for v as a function of t?
  - What if the object was moving at $t = 0$?
  - What if there was another force acting in the opposite direction?
- Frictional forces will depend on the fluid the object is moving through, its mass and how quickly it is moving.
- You will see examples of this in the practical and the exercises.

